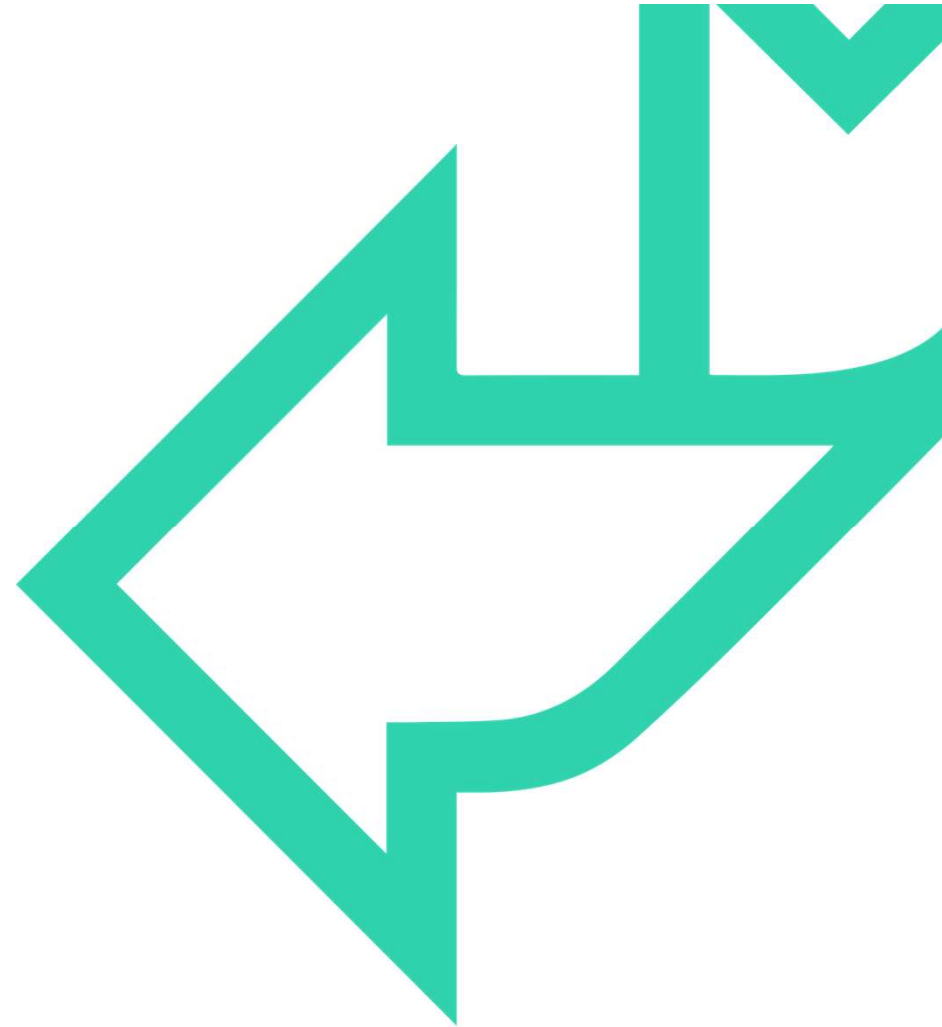




# Databases and SQL

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# **Introduction to Data Modeling**



# Timings

- Start 9.30 am
- Mid morning break 11.00am – 11.15am.
- Lunch 12.30pm – 1.30pm
- Mid afternoon break 3.00pm – 3.15pm.
- Finish 4.00pm
- Q&A 4.00pm - 4.30pm



# WHAT IS DATA?



| UNITED ELECTRICITY                                                |                   |                    |                                                                                                                         |                             |             |
|-------------------------------------------------------------------|-------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|
| Mr and Mrs Smith<br>21 Eastern Ave<br>Ealing<br>London<br>W13 100 |                   |                    | Bill period from 29 Dec 2007 to 3 Apr 2008<br>Tax point date 10 April 2008<br>Your customer account number 26367 234096 |                             |             |
| Meter number(s)<br>F03C29801<br>General Domestic                  |                   |                    | For any enquiry, please phone 0811 744 000                                                                              |                             |             |
| Reading last time                                                 | Reading this time | Tariff             | Units                                                                                                                   | Price of each unit in pence | Amount £ p  |
| 12234                                                             | 13423 E           | Unrestricted units | 1286                                                                                                                    |                             |             |
| Charges 29 Dec 2007 - 31 Dec 2007                                 |                   |                    |                                                                                                                         |                             |             |
| Standard energy                                                   |                   |                    | 22                                                                                                                      | 9.02                        | 1.98        |
| Discounted energy                                                 |                   |                    | 19                                                                                                                      | 8.55                        | 1.62        |
| Standard energy at 13.730p for 3 day(s)                           |                   |                    |                                                                                                                         |                             | 0.41        |
| Charges 1 Jan 2008 - 3 Apr 2008                                   |                   |                    |                                                                                                                         |                             |             |
| Standard energy                                                   |                   |                    | 705                                                                                                                     | 9.90                        | 69.80       |
| Discounted Energy                                                 |                   |                    | 540                                                                                                                     | 9.38                        | 50.65       |
| Standard charge at 15.070p for 93 day(s)                          |                   |                    |                                                                                                                         |                             | 14.02       |
| Monthly Direct Debit Discount VAT at 5.00% on charges of £131.04  |                   |                    |                                                                                                                         |                             | 7.44CR 6.55 |
| Total this invoice                                                |                   |                    |                                                                                                                         |                             | 137.68      |
| Balance from previous bill                                        |                   |                    |                                                                                                                         |                             | 40.17       |
| Payment received 20 January 2008                                  |                   |                    |                                                                                                                         |                             | 38.00CR     |
| Payment received 20 February 2008                                 |                   |                    |                                                                                                                         |                             | 38.00CR     |
| Payment received 20 March 2008                                    |                   |                    |                                                                                                                         |                             | 38.00CR     |
| DEBIT BUDGET ACCOUNT BALANCE CARRIED FORWARD                      |                   |                    |                                                                                                                         |                             | 83.78       |



# DATA ANALYSIS IS . .

... The process of examining an organisation's (or business system's) data independently of individual applications, so that a data structure can be developed which is capable of supporting all the organisation's (or business system's) func





# WHY DO WE NEED TO DO DATA ANALYSIS?

To determine the data requirements of the system  
(or organisation !)

To eliminate (or control) redundancy, so data is  
collected  
recorded  
and stored . . . once and once only

To enable high-level performance planning (at design  
stage)



# DATA MODELLING

Identify Entities (Tables) and their relationships

Entities can be physical things but also conceptual

Customer (Physical)

Transaction (Conceptual)

Always named in the singular

Draw diagrams that show the system



## Banking system:

loan

account

Customer

## Airline system:

aircraft

passenger

flight



# THE RELATIONAL DATABASE MODEL



| Rank | Title                                         | Director          | Worldwide gross | Year | Duration |
|------|-----------------------------------------------|-------------------|-----------------|------|----------|
| 1    | Avatar                                        | James Cameron     | \$2,783,165,628 | 2009 | 162      |
| 2    | Titanic                                       | James Cameron     | \$1,843,201,268 | 1997 | 194      |
| 3    | The Lord of the Rings: The Return of the King | Peter Jackson     | \$1,119,110,941 | 2003 | 201      |
| 4    | Pirates of the Caribbean: Dead Man's Chest    | Gore Verbinski    | \$1,066,179,725 | 2006 | 151      |
| 5    | Toy Story 3                                   | Lee Unkrich       | \$1,063,165,731 | 2010 | 103      |
| 6    | Alice in Wonderland                           | Tim Burton        | \$1,024,299,801 | 2010 | 108      |
| 7    | The Dark Knight                               | Christopher Nolan | \$1,001,921,825 | 2008 | 152      |
| 8    | Harry Potter and the Philosopher's Stone      | Chris Columbus    | \$974,733,550   | 2001 | 152      |
| 9    | Pirates of the Caribbean: At World's End      | Gore Verbinski    | \$960,996,492   | 2007 | 169      |
| 10   | Harry Potter and the Deathly Hallows: Part 1  | David Yates       | \$954,501,070   | 2010 | 146      |





# DVD COLLECTION - QUESTIONS

Give me a list of  
all films with  
runtime greater  
than 120 minutes

Give me a list  
of all the film  
titles in my  
collection

who directed  
Alice in  
Wonderland?

What films  
has Johnny  
Depp appeared  
in?



# DATABASE EXAMPLE TABLE 1

To An Online Film store: Films table

| Films        |                                               |                   |        |
|--------------|-----------------------------------------------|-------------------|--------|
| ProductCode  | Title                                         | Director          | Price  |
| TCF0046A8796 | Avatar                                        | James Cameron     | £11.93 |
| TCF00004CZDS | Titanic                                       | James Cameron     | £19.99 |
| NL000062V8V  | The Lord of the Rings: The Return of the King | Peter Jackson     | £6.79  |
| WD000PI3UNQ  | Pirates of the Caribbean: Dead Man's Chest    | Gore Verbinski    | £3.99  |
| WD002SNBIFS  | Toy Story 3                                   | Lee Unkrich       | £12.99 |
| WD0033WRUN2  | Alice in Wonderland                           | Tim Burton        | £6.00  |
| WB001CEE1W6  | The Dark Knight                               | Christopher Nolan | £6.99  |
| WB00288A1MY  | Harry Potter and the Philosopher's Stone      | Chris Columbus    | £4.48  |
| WD002Y2K2NM  | Pirates of the Caribbean: At World's End      | Gore Verbinski    | £3.99  |
| WB003U9VLKG  | Harry Potter and the Deathly Hallows: Part 1  | David Yates       | £14.99 |



## DATABASE EXAMPLE TABLE 2

### An on-line film store: BasketItems table

3 Customers

| Basket Items   |              |
|----------------|--------------|
| Shopper Number | ProductCode  |
| SH295870       | WD002SNBIFS  |
| SH295870       | NL000062V8V  |
| SH295870       | WB003U9VLKG  |
| SH295870       | WB00288A1MY  |
| SH295871       | TCF0046A8796 |
| SH295871       | WB00288A1MY  |
| SH295871       | WB001CEE1W6  |
| SH295872       | WD000PI3UNQ  |
| SH295872       | WD002Y2K2NM  |

Purchasing Different number of items

But what is an WD000PI3UNQ?



# DATA MODELLING PROCESS

## An on-line film store: BasketItems table

Produce the order for customer SH295872

Basket  
Items

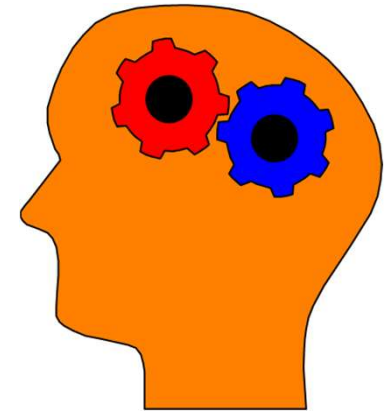
Films

Pirates of the Caribbean: Dead Man's Chest £3.99  
Pirates of the Caribbean: At World's End £3.99



# DATA MODELLING PROCESS

- Requirements gathering
- Analysis of enterprise information
- Identifying business entities
- Identifying business attributes
- Identifying business relationships



## The 3 Layer Model

User (or External) Views

Logical (or Conceptual) View

Physical (or Internal) View



# SOURCES OF INFORMATION FOR DATA MODELLING

**Avoid computer system documentation (initially!)  
Use a Top-down approach:**

- Management reports
- Front-end input





# A TYPICAL SOURCE

Order

Customer

| Sales Order Record     |              |                |                       |                   |               |
|------------------------|--------------|----------------|-----------------------|-------------------|---------------|
| Order No:              | Order Date:  | Order Value:   | Priority:             | Salesperson Name: |               |
| Customer No:           |              | Customer Name: |                       |                   |               |
| Invoice Address:       |              |                | Consignee Address:    |                   |               |
| ORDER DETAILS:         |              |                |                       |                   |               |
| Item No.               | Product Code | Description    | Qty                   | Price             | Delivery Date |
| 01                     |              |                |                       |                   |               |
| 02                     |              |                |                       |                   |               |
| 03                     |              |                |                       |                   |               |
| 04                     |              |                |                       |                   |               |
| 05                     |              |                |                       |                   |               |
| 06                     |              |                |                       |                   |               |
| 07                     |              |                |                       |                   |               |
| 08                     |              |                |                       |                   |               |
| 09                     |              |                |                       |                   |               |
| 10                     |              |                |                       |                   |               |
| Delivery Instructions: |              |                | Special Instructions: |                   |               |

Product

Order Item



# FROM ANALYSIS TO DATA MODEL...

